

Critical Incident Technique and Gig-Economy Work (Deliveroo): Working with and Challenging Assumptions around Algorithms

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ABSTRACT

Decision-making algorithms can be obscure and fast-moving. This is especially the case in the context of the algorithm that mediates the work of Deliveroo riders. Forming a critical part of the food delivery platform, the algorithm's obscurity and shifting nature is a part of its design. In this paper we argue that adapting usability techniques like the Critical Incident Technique (CIT) may provide one way to better understand algorithms and platform work. Though there are many methods to understand algorithms like this, asking people about negative or positive interactions with them and what they think provoked them can produce fruitful avenues for HCI research into the impacts of platforms on gig-economy work. We argue that despite the results being an assumption, assumptions from the algorithmically managed are interesting materials to challenge the researchers' own assumptions about their context, and to, therefore, better scope out contexts and iterate future research.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; • **Applied computing** → *Transportation*.

KEYWORDS

Critical Incident Technique, gig-economy, algorithms, incidents, couriers, workers

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1 INTRODUCTION

Daily life is increasingly mediated by algorithms, and suggestions have been made about the different ways that they can be studied [10, 22]. In this paper, we outline how we approached the problem

of understanding Frank, Deliveroo's algorithm [9], and the ways that it shapes and mediates the work of food delivery workers.¹ In doing this, we aim to provoke discussion at CHI around the adaptation of established usability methods for new contexts.

Usability testing has produced a wide variety of methods, including diary-based longitudinal user reporting and user-based laboratory testing [5]. Normally, these methods are intended to evaluate a technology or service that is in production with users during different moments of its development [30]. The Critical Incident Technique (CIT) is one such method and has been used in User Interaction (UI) design [26]. It seeks to better understand the positive and negative interactions that users have with a product, as well as what they believe caused these incidents to allow their developers to improve upon it.

We adapted this technique and made use of it as a survey device to better understand rider incidents with Frank. CIT helped us to address a specific problem in our work, accessing a hard-to-reach community during the pandemic when our usual methods were out of scope. But when CIT is mobilised in usability testing, users – usually – have a clearer sense of what caused an incident. For instance, clicking a particular part of an interface did not produce the intended outcome. However, when CIT is mobilised in reference to a hidden and remote algorithm, as in our case, the algorithmically managed must make assumptions about what they think caused the incidents reported. This, sometimes, has little to do with their interactions with it because, though the work of riders is mediated by an algorithm, many of the actions that they were undertaking were out-in-the-world (i.e., collecting and delivering food) – meaning that the causes can be wide-ranging, and include a variety of different actors (e.g., customers, restaurants, other city users). In this short paper, we argue that adapting usability testing methods like CIT to better understand algorithms' effects on activities is a fruitful avenue for future investigations, and to scope out research contexts. Our aim in doing this is to provoke a discussion around reusing methods like CIT for new contexts, like understanding the algorithms that mediate platform work, but also around the role of assumptions in research itself.

2 RELATED WORK

2.1 Platform Work and Algorithms

There is a growing number of gig-economy workers in the UK, with these working across many different contexts from delivery [4] to

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¹Deliveroo terms their workers as riders. This distinction matters given the contested contractual status between employees and independent contractors that exists in gig economy work [28].

freelance creative work [8]. Typically, such workers are paid per task (in an arrangement sometimes known as piecework [3]) and receive work on an '(in)dependent contractor' basis [32] with platforms (e.g., Deliveroo, Uber, Upwork) apportioning work to those capable of supplying it [4]. Data is collected through these platforms and feeds into their algorithmic management [23, 24] with hidden processes deciding when and which workers are offered 'gigs' [6]. Whilst these algorithms minimise the costs of middle-management for the platform, Kellogg et al. [20] argue that positivist views of algorithms at work are problematic due to the levels of control that they have over workers (e.g., direction, evaluation, discipline).

2.2 CIT

CIT has a longstanding history in usability testing [17] but it was initially introduced by Flanagan [14] for other contexts, including counselling and psychotherapy. Flanagan [14] describes the method as a set of procedures for collecting direct observation of human behaviours for solving practical problems and developing (psychological) principles. This set of procedures, however, is not rigid, and this is reflected in its flexible application by the HCI community. In general, it is a method that informs developers of how users experience a product at 'critical' points that help designers optimise it for future users [26]. It asks users to retrospectively describe a situation about a product or service that, in their view, either ended well or poorly.

Hartson et al., [17] used it as a form of remote evaluation tool, when confronted with the challenge of testing the 'network' and its many distributed and remote users. Mackay & Fayard [25] suggested that CIT could be used as an exercise into which video could be integrated, in their tutorial of how researchers could use video to support a range of participatory design activities. Others, like van Dijk et al., [34] and Dove et al., [11] have combined the method (or parts of it) with others, when studying their contexts of multi-channel consumer behaviour in e-commerce, and user experience when interacting with a marketing company, respectively. Meanwhile, Akers et al., [1] mobilise user-reported CIT as a tool to compare with the results of an algorithm developed to identify critical incidents through usage log data. Finally, it has been used – as with our context – as inspiration; adapted to meet a specific situation at hand – as recommended by Flanagan [14]. Wagner & Mackay [35] use a variation of CIT in their five-week study of print jobs, exploring the details of each printout. Morrison et al., [27] adapt it further, conducting 10 online surveys assessing internet demographics and usage patterns, making use of what they term the 'Significance Question' to mobilise aspects of Flanagan's [14] procedures in exploring what activities significantly impact people's decisions and actions.

Positive and negative incidents tend to be analysed and reported separately, and the findings are used to make recommendations to improve future outcomes, or the ease-of-use of a technology or service. Having explained CIT and its uses in the CHI community, we now explain the context and project in which we applied it.

3 DELIVEROO, FLIPGIG, AND CIT

Deliveroo, the food delivery platform, was launched and founded by Will Shu and Greg Orłowski in London in 2013 [9]. It matches

moments of restaurant and takeaway food demand with riders capable of supplying it. Frank allocates work to riders, calculating which is best placed to do each gig, as well as predetermining the fees for completing the work. In other words, the algorithm has power in deciding who works, when, and for how much. This power is further compounded by the independent status of the riders, as if gigs are not allocated to a rider, they are not paid. Such an arrangement has produced diverse experiences of pay, job allocation, as well as suspicions around how Frank operates [4].

This arrangement has also attracted multi-disciplinary research attention, not just on the experience of the work [7, 15, 31, 33], and the role of the algorithm within it [16, 18, 38], but also of the dynamics of the rider community's resistance in response [7, 36]. This body of work paints the experience of being a Deliveroo rider as one of risk, precarity, and unpredictability – with Frank and the lack of worker protections being key influences in this experience.

As noted by the work above, the operations of Frank are truly obscure, and – in addition – can change with little to no notice through 'push updates'. Deliveroo riders are, too, a hard-to-reach community: fast-moving and under pressure to maximise their working time. In working for a digital platform there is also no formal site to approach and recruit them through, rendering them somewhat invisible outside of working hours.

FlipGig has been exploring how digital technologies can be used to target issues of fairness and sustainability in gig-economy work, like Deliveroo. It began in April 2020, just as the COVID-19 pandemic began to take hold in the United Kingdom and this brought new research challenges to an already complex context. We were not allowed to directly approach riders, finding ourselves under strict instructions to avoid face-to-face research methods. New methods were, therefore, required to engage with this community, and Frank's role in it. It was in this context that we sought out established surveying methods and came across CIT, deciding to use it to understand how Frank worked in practice, and the assumptions it produced in riders.

Given the origins of the method, it should not be surprising that some adaptation was required (see Table 1). We are not the developers of Frank, nor were we looking for ways to improve it. Instead, we were interested in how riders believed that it was working, and what effects they felt it was having upon the gigs they were carrying out, and their work more broadly.

As can be seen, we removed the language of 'users', looking to situate the experience of the 'incidents' within the context that they were happening. Similarly, we changed 'incidents' to 'experiences', recognising that this might improve the intelligibility of the question to anyone undertaking the survey outside of an environment for usability testing. The language of experience was used as it is commonly known to refer to an event or occurrence which has left an impression. We combined questions one and two, with a more general open question of 'tell us', acknowledging that the behaviours of the riders and their actions (as users of the platform) could be very similar in this context. Because we wanted to learn as much as possible through this survey, we also added a final question, allowing riders to report more generally around the incident that they were reporting.

Despite the changes made to the questions, the general intention of the method, however, remained the same: collecting incidents

Table 1: Questions (as outlined by Martin and Hanington [26]) compared to our adaptations

Question	CIT as described by Martin and Hanington	Our adapted questions
<i>Q1. The incident cause</i>	What were the events leading up to the critical incident?	Tell us about your experience and what you think caused it?
<i>Q2. User actions</i>	What were the behaviors that took place during the incident?	(combined with Q1)
<i>Q3. User sentiment</i>	How did the user feel during the incident, and afterward?	How did this make you feel during the experience and afterwards?
<i>Q4. Incident outcome</i>	Did the user change how s/he behaved after the incident?	Has this led to a change in how you work? Or what might happen if you don't change how you work?
<i>Q5. Ideal outcome</i>	If behaviors change, what are other possible future outcomes?	What would have been a better outcome?
<i>Q6. Anything else</i>	n/a	Is there anything else you wish to tell us about this experience?

(both positive and negative), assumptions around the cause, as well as better outcomes. Given the project's interest in fairness for gig-economy workers, collecting 'better outcomes' worked towards trying to understand what riders felt fairness in their work would look like – rather than gathering responses to help make improvements to a product.

After developing the survey, we put it online and recruited through known couriers, asking them to share within the WhatsApp groups common in this community. This resulted in 26 responses. Responses were coded for where within the decision-making process of the algorithm the incident emerged from (e.g., login, shift allocation etc.), and the actors mentioned. Where multiple factors were mentioned, these were coded across categories, resulting in more codes than reports: 12 codes, 58 coded responses across 26 surveys.

The incidents collected were entirely negative. Though the survey emphasised our interest in incidents of all kinds, negative experiences are sometimes remembered over the positives in the long-term [21]. In addition to this, those with negative experiences may have been more motivated to report them, than those with only positives. As CIT is normally conducted with a range of people who have *just* used a product, they are more likely to report some of the positives too. However, our interest in fairness for couriers meant that this was not necessarily a limitation of the method for us as the results worked to show what riders felt was unfair about how the platform operated. For other contexts, however, this limitation may challenge a more holistic understanding of algorithms and their impacts.

Other limitations did exist, however. For one, snowball recruitment of participants may have skewed the results somewhat, in that those that responded were already connected through shared WhatsApp groups, who could have synergistic ideas about how the work worked. Secondly, the adaptations we made to the questions may feel – to some – a bit generic as we strove not to over tailor them to our problem domain to avoid biasing the respondents. Despite these limitations, as we will now demonstrate, the results of applying CIT to our context proved valuable for the project's understanding of Deliveroo work.

3.1 Results and Discussion

In what follows, we discuss the results from this work, organising this around the two themes of: 1) challenging researcher assumptions, and 2) rider assumptions.

3.2 Challenging Researcher Assumptions: Opening up our focus

What resulted from this method surprised us. Firstly, it challenged our dominant assumption – one that even underpinned the funding application for the project – that it was the algorithm itself that predominantly produced unfairness in the work.

Fig 1 demonstrates that most of the codes were for incidents occurring 'outside' of the algorithm (36 vs. 22).

"I was very surprised that on my first day on the platform that [there] was no induction on the role itself, where to go if [there] was a problem with the equipment especially [given that] 95% of the time you're on your own [...]" (Participant 6)

"I have been kept waiting for really long times at restaurants – sometimes half an hour or longer [...]" (Participant 13)

These involved the (lack of) training for the job, collecting from (slow) restaurants, as well as rider support. Though training and rider support are facets of the Deliveroo platform, they are human-mediated factors of the platform, rather than the details of Frank itself. Even when incidents occurred 'inside' the algorithm, these were often about how other actors, like customers, influenced the algorithm in ways that were not favourable to the rider.

"I had a customer enter an incorrect address which didn't exist [...] he asked me to deliver to an address outside my zone [...] I told him I was cancelling the order. I was really frustrated by the loss of time as I was not paid for this, and Deliveroo made me return the food to the restaurant" (Participant 21)

Customers who lied about their address or entered it incorrectly to access food not available in their area, led to a reduction or loss of fees for the job (predetermined by location) not reflecting the

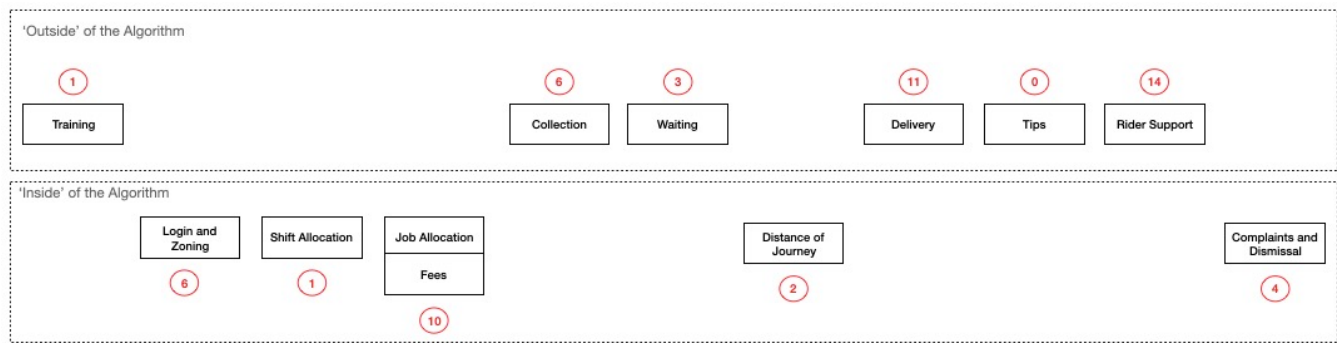


Figure 1: An overview of the job process and of where the critical incidents emerged within it.

extra distance and work (i.e., phoning) involved in finding and delivering to them. Rider support, however, was the most frequent code (14), though this was usually related to its response to incidents occurring 'outside' of the algorithm and with other stakeholders.

"[...] The address of the last order was not correct, so I tried to reach the customer by telephone. It was impossible [...] I called the rider support, and I explained the situation. They said that everything was fine, so I gave the pizza to a homeless [person] on the street and I went home. Around 20 minutes later, Deliveroo called me back asking me to finish the order because they were able to contact the customer [...] next day they sent me an email warning me that behaviour like this might result in dismissal" (Participant 16)

Four riders reported threats of dismissal (and in one case being dismissed) following negative incidents with others, including unrelated city users (e.g., attempts to steal vehicle, drunks).

In short, the insights gained from applying the critical incident technique to Frank demonstrated how experiences of unfairness were often being produced by interactions of riders with other stakeholders outside of the algorithm. This is — of course — not to shift the responsibility of the working conditions away from Deliveroo. There are certainly things that could be improved by the platform like improving location services to ensure that customers cannot lie about their location or penalising consistently slower restaurants. However, CIT did begin to open the focus of the research into the consideration of other actors (e.g., restaurants, customers, and other city users) and their impact upon the conditions and resulting pay of food delivery work. Though many incidents were from 'causes' out-in-the-world that riders were working in, the way that the platform — and at times, algorithm — responded to such incidents was often perceived as inadequate.

3.3 Working with Rider Assumptions

These insights went on to shape later research in the project, changing the focus of interviews and the details of the interview probes used within them. Despite having been involved in logistics research previously, this method sensitized the project in ways that the previous work had not. We now understood that the algorithm and Deliveroo, though not always responsive to the issues that food

delivery workers faced in their work, was not always the cause of these. This expanded our thoughts of how digital technologies could help to improve working conditions for gig-economy workers.

Before applying CIT to our context, we recognised that what would result would be assumptions from the riders about how the algorithm might be working. Yet faced with a new research context and limitations to how we could carry out our work, it was decided that these assumptions could still provide useful insights — and they did. Riders are experts of their own domain and they do have more knowledge of the algorithms and context under investigation than we did. In certain cases, we had riders reporting the same incident, but suggesting different causes for it. For instance, multiple causes were associated with slow restaurant times.

"I believe [...] Deliveroo over hires new riders [...] to compensate for this [and...] to keep riders engaged and busy during a shift and discourage us from logging off. They assign orders long before they will be ready to collect" (Participant 14)

"Restaurants are disrespectful and can have huge waiting times because they prioritise their own customers over third party apps" (Participant 17)

A multiplicity of causes, rather than being a negative, provided a wider field for research, and thinking through food delivery working activities. It also produced conversations within the project around the status of assumptions, and how even in other and more established methods, researchers are usually working with assumptions of all kinds, not only from participants who may be asked to hypothesise around different situations (e.g., Why something has happened?) but from themselves, as well, about the context under investigation. We believe, however, that this is particularly the case when researching the algorithms that mediate platform work.

Others have made similar points in reference to assumptions and the study of algorithms, with this particularly being the case when there is an absence of knowledge about how they work. DeVito et al. [10] for instance, use interview data to explore how social media users form folk theories around algorithmic feeds, and how their assumptions impact self-presentation processes online. Other HCI work has sought, similarly, to understand such folk theories in algorithmically managed contexts, with assumptions and self-beliefs also playing roles in this literature [2, 12, 13, 29]. Relatedly, Jarrahi

and Sutherland [19] demonstrated how Upwork users developed algorithmic competencies through sense-making activities when dealing with algorithmic management from platforms. Moreover, Wiener et al. [37] presented theoretical arguments and empirical evidence to counter the generally negative rhetoric that exists around algorithmic control, arguing that there are opposing effects, with workers having some positive associations with elements of it.

Our argument, though originating from the same starting point, differs somewhat. We suggest that adapting usability techniques, like CIT, which ask 'users' to evaluate (negative/positive) moments with algorithms, whilst asking for them to assume a cause, could act as a useful first scoping step in algorithmically managed contexts. Doing so allows researchers to fine-tune their focus and interview questions, before undertaking more time-consuming methods. This was especially important in our context given that it led for us to de-centre the algorithm all together from the preceding investigations, and in our understandings of the work of Deliveroo riders. Through the application of this method, we learnt that though Frank does feature heavily in the experience of these riders, there are other crucial elements that contribute to the production of unfairness in their work.

4 CONCLUSION

In this short paper, we have argued for the adaptation of established methods from usability testing to better understand algorithms and platform work. We adapted CIT and found that though what resulted was riders' assumptions about how Frank worked, these were fruitful avenues in our project, challenging our own assumptions about the work and allowing for further iteration of the project's focus prior to undertaking interviews with food delivery couriers.

Though scoping methods are not new, and neither is the application of CIT, we believe that the combination is novel in the context of understanding algorithms that mediate platform work. This is a challenging context to research, particularly when such algorithms are designed to be deliberately obscure. We could not speak with Frank's developers, nor could we access the source code. Frank can also change, and sometimes it is unclear if it has done so [6]. However, the complexity of some algorithms should not distract researchers from the main goal of their work. In our case this was to understand how food delivery work operated, and how digital technology could be used to improve that work. Though much has been written about Deliveroo's Frank, riders reported that it was not the algorithm itself that caused many of the problems, but rather issues faced on-the-ground when delivering meals, and the platforms response to this. Using CIT in this context, therefore, challenged our own perceptions about the struggles of Deliveroo work, opening our focus and better iterating later phases of the research.

We believe that non-developers should embrace usability techniques for contexts other than usability, as they provide useful scoping methods that are quick, whilst asking users to make assumptions or share their beliefs about what they think caused an incident. Given the role of self-beliefs and assumptions (i.e., folk theories) in understanding how users interact with and are influenced by algorithms, this is particularly useful in this context. Future work might consider how to better tailor established surveying

questions, like CIT, to algorithms and the contexts in which they are used. However, even in other contexts, participant assumptions have power in challenging the researchers' own assumptions when approaching their contexts of study. Assumptions from those who are experts in their own domain can provide useful details and variety to better plan later stages of the project.

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